What is Fetch API

→ JS interface used for making HTTP request to servers.

→ Return Promises

→ Used to get data

QS

Basic Method / then.catch(Promise chain)

```
fetch('https://api.example.com/data')
  .then(respons => response.json())
  .then(data => console.log(data))
  .catch(error => console.error('Error', err
```

Modern Method - async/await

QS

```
async function fetchData(){
  try{
```

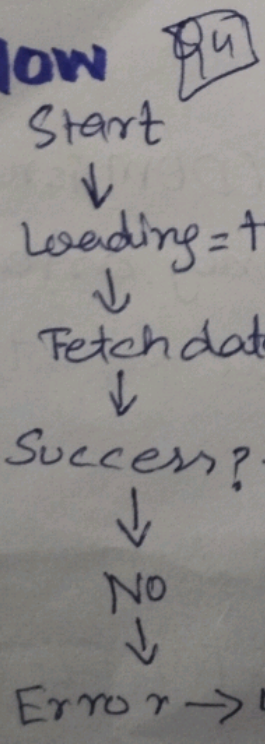
```
const response = await fetch(.....);
const data = await response.json();
console.log(data);
}catch(error){
  console.error("Error", error);
}
}
```

Handling Loading, Sucess, Error State

→ In react we track three states when fetching data:

```
const [data, setData] = useState(null);
const [loading,setLoading] = usestate(true);
const [error, setError] = useState(null);
```

Flow



→ Don't call fetch in component body Q1
→ call it in useEffect

```
function Component(){
  const [data, setData] = useState(null);
    useEffect(() => {
  fetch(------)
  .then(res => res.json())
  .then(data => setData(data));
  }, []);
  }
```

Why useEffect? Q3

→ Prevents infinit re-renders
→ Runs after component mount (DOM is ready)
→ Can be controlled with dependency array
→ Support cleanup function (cancel request)

Why do we use Fetch API?

→ Get data from servers/APIs
→ Send data to servers (POST, PUT, DELETE)

→ Update React App with external data
→ Create dynamic, data, driven apps

Fetch

→ Built in Javascript
→ Manual JSON parsing required
→ No request cancellation
→ less browser support
→ Lightweight

```
fetch(url)
  .then(res => res.json())
  .then(data => setData(data));
```

Axios

→ External library
→ Auto JSON parsing
→ Built-in request cancellation
→ Better Browser support
→ More features, slightly heavier

```
axios.get(url)
  .then(res => console.log(res.data));
```

What happens if you forget the dependency array in useEffect when fetching data

Fetch will run on every render, causing
→ Infinite loop (fetch → update state → re-render - fetch)
→ Performance issues
→ App crash
→ Wasted API calls

Always use depency Arry
→ run once on mount - []
→ run when variable change - [variable]